

Gregory Levy

gregory.levy@economics.ox.ac.uk — gregorylevy.github.io

Education

University of Oxford (UK)

DPhil in Economics	2028 (expected)
<i>Econometrics, Political Economy, Foerster Lab for AI Research (FLAIR)</i>	
<i>Associate Researcher, London School of Economics (2026–present)</i>	
MPhil in Economics (with Distinction)	2025

University of Queensland (Australia)

Honours Degree in Economics	2022
<i>Synergies Consulting Prize for Best Honours Thesis</i>	
Bachelor of Mathematics	2021
Bachelor of Economics	2021

Selected Research Projects

Econometrics and Machine Learning Theory

A Framework for Optimal High-Dimensional Variable Selection:

Introduces the Manifold-adaptive Entropic Shrinkage Algorithm (MESA) for selecting relevant variables in high-dimensional datasets. Develops a framework that identifies optimal regularisation method according to the data structure and researcher's objective. Demonstrates improved stability and out-of-sample performance on large-scale political text data.

Neural Networks as Optimal Continuous Piecewise-Linear Approximators:

Shows that standard neural networks can be understood as efficient linear approximators of complex functions, and identifies how their performance depends on underlying features of the data. Combines theory and simulation to explain when and why neural networks work well, and develops tools to guide the choice of model architecture in practice. Joint work with Jakob Foerster (Oxford).

Empirical Political Economy

Measuring Democratic Responsiveness across Competing Principals:

Studies how elected representatives respond to competing pressures from voters, donors, and political parties. Develops a unified Bayesian framework to measure these influences across the UK, US, and Australia, providing new evidence on the sources of political representation and accountability.

Political Uncertainty Classification and Forecasting:

Builds a large-scale system to measure political instability from historical newspaper text (1800–present). Uses modern language models to classify and track political risk over time, generating new data for research and policy analysis. With Michael Callen (LSE), Jonathan Weigel (UC Berkeley) and Noam Yuchtman (LSE). [Forthcoming in *Handbook of Political Economy* (Acemoglu & Robinson, eds.).]

Conferences and Workshops

Presented at inaugural UK-GEN Workshop (Warwick, 2026)

Presented at ICMS ‘AI and Economics’ Workshop (Edinburgh, 2026)

Attended Plurality Institute ‘LLMs and Public Discourse’ Workshop (Berkeley, 2025)

Attended Machine Learning in Economics Summer Institute (University of Chicago, 2024)

Presented at Foundations of Utility and Risk (FUR) Conference (Brisbane, 2024)

Presented at Learning, Evolution, and Games (LEG) Conference (Brisbane, 2024)

Teaching

Out-of-College Tutor in Economics — Magdalen College, Oxford 2026 – present
Undergraduate Economics Thesis Supervisor.

Stipendiary Lecturer in Economics — Magdalen College, Oxford 2024 – 2025
Preliminary Microeconomics (MT 2024); Core Microeconomics (TT 2025, MT 2025).
Founded and led extracurricular Machine Learning and Economics discussion group for Magdalen undergraduates.

Teaching Assistant — University of Oxford Summer Schools 2025 – 2026
Geeconomics Uncovered: Theory Meets Evidence (June 2026).
AI and Machine Learning Pioneers Summer School, Saïd Business School (August 2025).

Academic Tutor — University of Queensland 2017 – 2022
Tutorials and workshops in mathematics, economics, statistics, chemistry, and scientific modelling.

Private Tutor and Educational Speaker 2014 – present
Approximately 200 students across a broad range of subjects, secondary school through graduate level.

Service and Leadership

Co-founder, UK Graduate Econometrics Network (UK-GEN) 2026 – present
Founding committee member of organisation connecting British doctoral researchers in econometrics. Co-organised inaugural workshop (Warwick, June 2026).

Organiser, Oxford Machine Learning and Economics Reading Group 2024 – present
Assistant organiser of interdisciplinary reading and discussion group convened with Professor Maximilian Kasy, bringing together faculty and research students across economics, statistics, computer science, and philosophy.

Honours, Scholarships, and Awards

Magdalen College Graduate Scholarship (Oxford)	2023
Synergies Consulting Prize for Best Honours Thesis (UQ)	2022
Dean’s Honour Roll (UQ)	2021